Device Overview

Introducing NCD's Long Range Industrial IoT Wireless CO2, Temperature, Humidity, and Particulate Matter Sensor for air quality monitoring. It features a range of up to 2 miles using wireless mesh networking, transmitting CO2 concentration, particulate size, mass, temperature, and humidity data at customizable intervals.

The sensor uses CMOSens® Technology for precise CO2 measurement, along with NDIR technology for CO2 detection. It includes a best-in-class humidity and temperature sensor, which compensates for external heat sources without additional components. The dual-channel design ensures long-term stability in CO2 measurement.

The particulate matter sensor measures PM1.0, PM2.5, PM4.0, and PM10.0 particles, which can cause serious health issues like asthma and cardiovascular disease. It uses laser scattering technology for precise measurements and includes a fan for automatic sensor cleaning every 10 transmissions.

The temperature sensor operates from -25°C to 85°C (-13°F to 185°F), while the humidity sensor handles 0-100% RH. Powered by 6 AA batteries, the sensor offers up to 3 years of battery life, with external power options available on request.

Features

- Industrial Grade Sensor Mass Concentration Range 0-1000 µg/m^3
- Mass Concentration Resolution of 1 µg/m^3
- Mass Concentration Size Range PM1.0, PM2.5, PM4.0, PM10.0
- Number Concentration Range 0 to 3000 1/cm^3
- Number Concentration Size Range PM0.5, PM1.0, PM2.5, PM4.0, PM10.0
- Industrial Grade CO_2 Measurement Range 400 – 5'000 ppm
- CO2 Measurement Accuracy ± (60 ppm + + 5% of reading)(25 °C, 400 – 5'000 ppm)
- Temperature compensation of CO_2 sensor signal
- Fully calibrated and linearized
- Unique Long-Term Stability of up to 8 Years
- Laser-Based Scattering Principle and Advanced Algorithms
- Built-in Auto Cleaning Feature (internal fan) After Every 10 Transmissions
- Industrial Grade Sensor with 0 to 90% RH / -25°C [-13°F] to 85°C [185°F])
- Resolution 0.4% RH/0.1°C Accuracy 6% RH/ 0.8°C
- Encrypted Wireless Communications up to 2 Mile Range in open environments

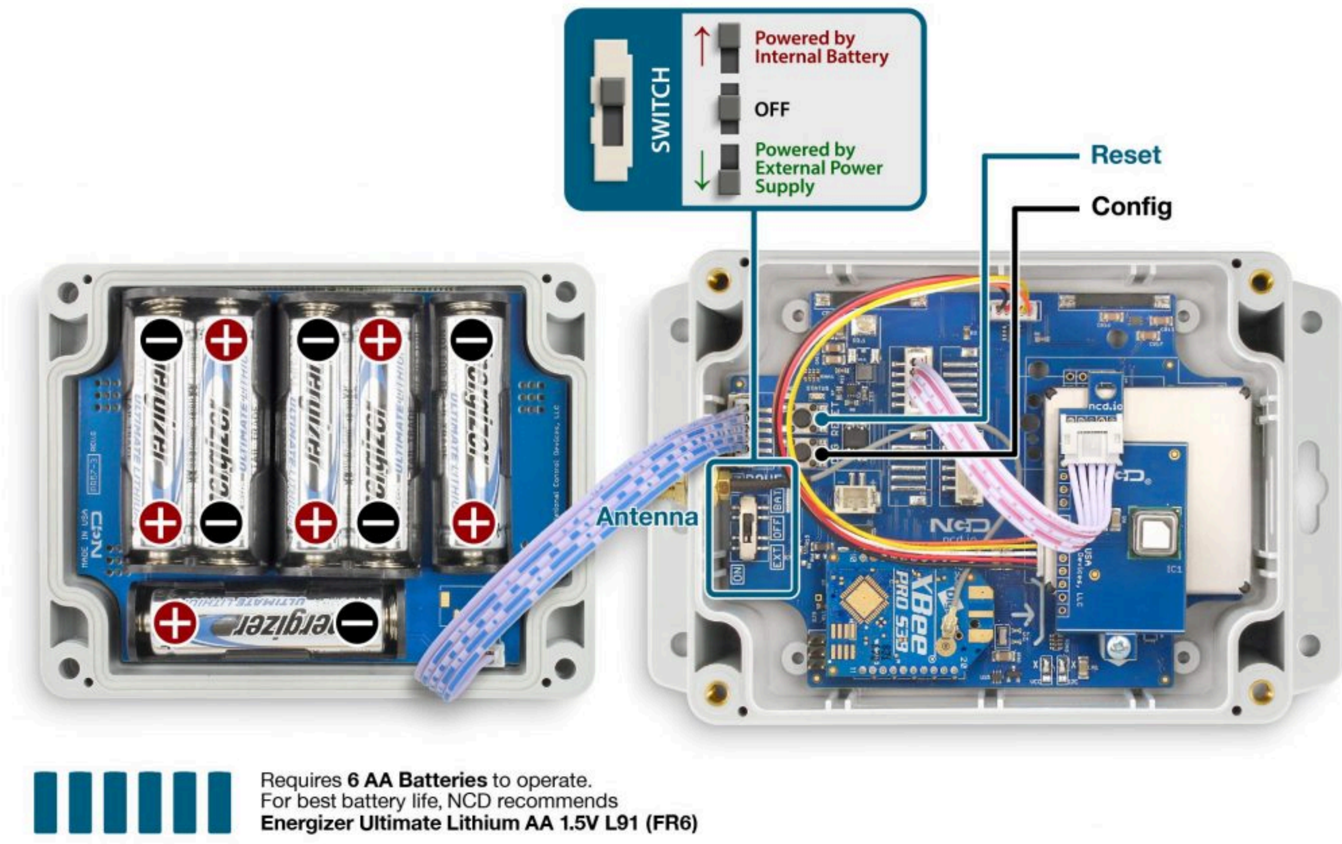

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Figure 1: Device Overview

Status LED

The Status LED is used to indicate errors or other sensor diagnostics.

LED blinks once – Message was sent successfully and no error in last sensor read as well in data transmission.

LED blinks twice and then one more time – Message was sent successfully but there was an error in last sensor read.

LED blinks thrice – MCU is having issue communicating with the radio module.

Sensor Specifications

Specifications	Minimum	Nominal	Maximum	Notes
Width		3.54"		
Length		4.52"		

Specifications	Minimum	Nominal	Maximum	Notes
Height		2.16"		
Enclosure Rating		IP65, NEMA 1,2,4,4X,12,13, UL-508		
Mounting		1/4 NPT 1/4-28 UNF Magnet Mount		
Temperature Rating	-40° C	23° C	85° C	Component Rating
Tested Temperature	0° C	23° C	40° C	As Tested by NCD Staff
Probe Temperature	-40° C	23° C	105° C	As Tested by NCD Staff
Sampling Rate	30 seconds		20000 seconds	
Mass Concentration range	0 $\mu g/m^3$		1000 $\mu g/m^3$	
Mass Concentration resolution		1 $\mu g/m^3$		
Mass Concentration precision		±10 %		
Mass Concentration size range				PM1.0, PM2.5, PM4.0, PM10.0
Number Concentration range	0 $1/cm^3$		3000 $1/cm^3$	
Number Concentration size range				PM0.5, PM1.0, PM2.5, PM4.0, PM10.0
Temperature range	-25°C [-13°F]		85°C [185°F]	
Temperature resolution		0.025°C		
Temperature accuracy		0.3°C		
Relative Humidity range	0 %		100 %	
Relative Humidity resolution		0.04%		

Specifications	Minimum	Nominal	Maximum	Notes
Relative Humidity accuracy		2%		
CO2 Measurement range	400 ppm		5000 ppm	
CO2 Measurement accuracy		60 ppm + + 5% of reading		

Printed Circuit Board Specifications

Parameter	Minimum	Nominal	Maximum	Notes
Trace Width	0.007 inch	0.012 inch	0.25 inch	Trace Width depends on the Trace Type
Layer Count		2L - Rigid		Top and Bottom Layer
Material Type	FR-4 TG170	FR-4 TG170	FR-4 TG180	
Surface Finish		ENIG 2u"		
IPC Standard		IPC CLASS 2		
Finished Copper Foil		1.0/1.0 OZ		
Finished Thickness		0.062 inc		
Fine line <4.0/4.0mil		No		
Blind & Burried Vias		No		
Non-Conductive Resin		No		
Conductive Resin		No		

Mechanical Drawing

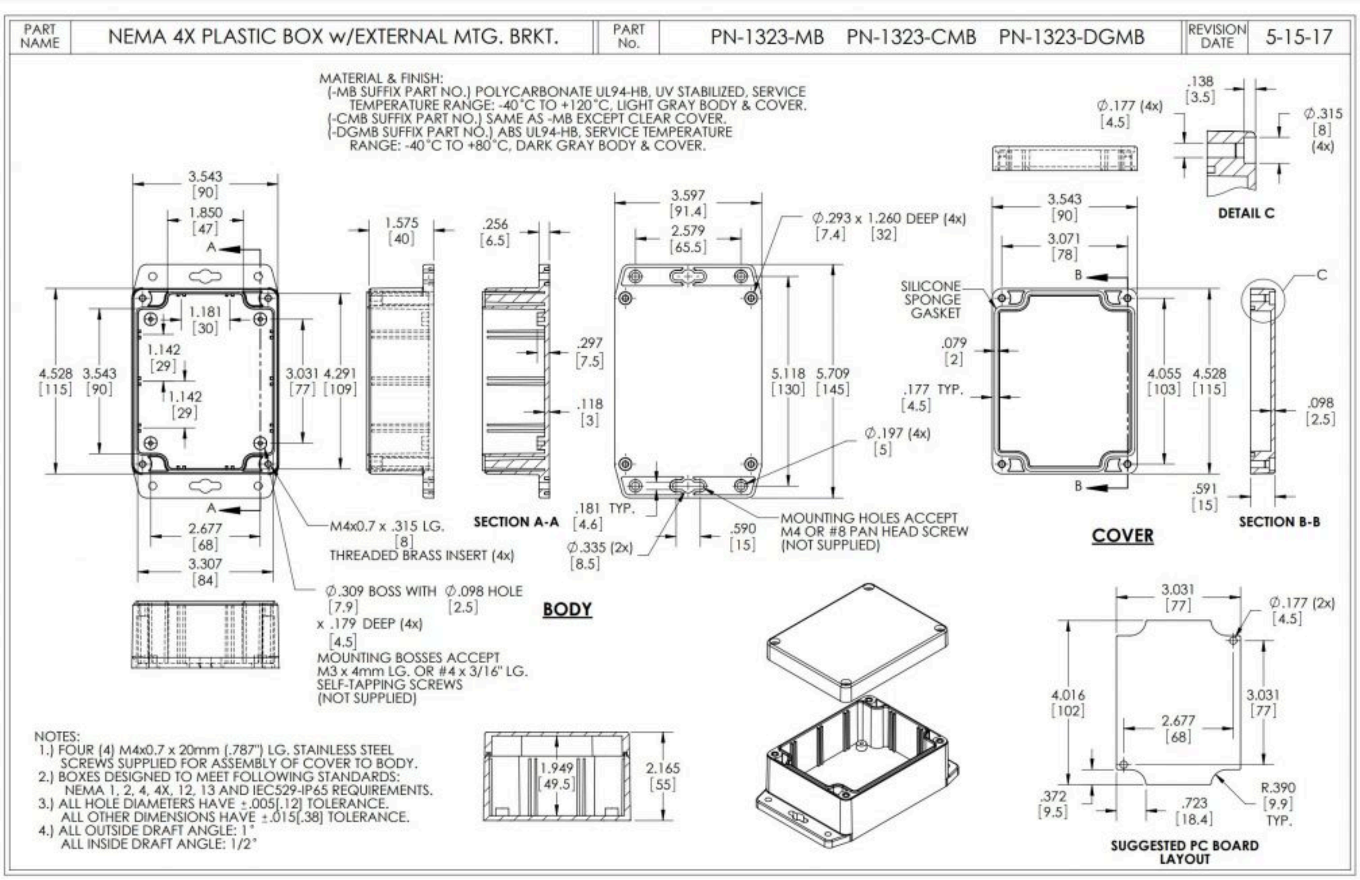

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Figure 3: Mechanical Drawing

RF Module Specifications

Parameter	868MHz	900MHz	2.4GHz
Frequency Band	863 MHz to 870 MHz	902 to 928 MHz	ISM 2.4 GHz
Transmitter Power	Up to 13 dBm ERP	Up to 24 dBm	Up to 8 dBm
Receiver Sensitivity	-106 dBm at 80 Kbps	-101 dBm at 200 Kbps	-103 dBm at 250 Kbps
Range (dense urban)	~1000ft	~1000ft	~300ft
Range (line of sight)	~2 miles	~2 miles	~1 mile

Parameter	868MHz	900MHz	2.4GHz
Data Rate	80 Kbps	200 Kbps	250 Kbps
Networking Protocol	Digi XBee® DigiMesh®	Digi XBee® DigiMesh®	Digi XBee® DigiMesh®
Encryption	128-bit AES	128-bit AES	128-bit
Reliable Packet Delivery	Retries/acknowledgements	Retries/acknowledgements	Retries/acknowledgements
IDs	PAN ID and addresses, cluster IDs and endpoints (optional)	PAN ID and addresses, cluster IDs and endpoints (optional)	PAN ID and addresses, cluster IDs and endpoints (optional)
Certification	CE/RED, ROHS Compliant	FCC (America), IC (Canada), C-Tick (Australia), Anatel (Brazil), IDA (Singapore)	FCC (America), IC (Canada), RCM (Australia), Anatel (Brazil), Teleck MIC (Japan), KCC (South Korea)

Power Requirements and Expected Battery Life

Power Requirements

This Sensor has two power options:

1. Powered by AA batteries (NCD recommends Energizer L91 batteries)
2. External power supply (6-18 VDC) Current Requirement 250 mA

Note! ×

The L91 Batteries are non-rechargeable Li-Ion batteries.

While changing the batteries:

1. Turn off the sensor
2. Only use new L91 batteries
3. Checkout the polarity marking on the battery holder and batteries
4. Replace all six batteries with new L91 batteries. Do not mix old and new batteries
5. Batteries should not be replaced in fire-risky areas
6. Do NOT install rechargeable batteries

Expected Battery Life

Specifications	Minimum	Nominal	Maximum	Notes
Batteries	2	6	6	May be Powered by 2 or 4 AA Batteries
Battery Life 1 TPD			5 Years	TPD Transmissions per Day
Battery Life 12 TPD			2 Years	TPD Transmissions per Day
Battery Life 24 TPD			1 Year	TPD Transmissions per Day

Note! ×

The Truth About Battery Life Under the best of circumstances, the best non-rechargeable batteries commonly available today are limited to a 10 year non-working shelf life in a room temperature environment. Factors such as actual usage, temperature, and humidity will impact the working life. Be wary of any battery claims in excess of 10 years, as this would only apply to the most exotic and expensive batteries that are not commonly available. Also note that most battery chemistries are not rated for use in extreme temperatures. NCD only uses the best Non-Rechargeable Lithium batteries available today, which are also rated for use in extreme temperatures and have been tested by our customers in light radioactive environments. Lithium batteries offer a 10 year maximum expected shelf life due to limitations of battery technology. NCD will never rate sensor life beyond the rated shelf life of the best batteries available today, which is currently 10 years.

Store and Documentation

Head to the NCD store if you want to purchase the
Industrial IoT Wireless Air Quality CO2 Temperature Humidity Particulate Matter Sensor (https://store.ncd.io/product/industrial-iot-wireless-air-quality-co2-temperature-humidity-particulate-matter-sensor/).
If you already have the device and would like more information into its basic and advanced operations check out the Quick Start Guide (https://ncd.io/blog/industrial-iot-wireless-air-quality-co2-temperature-humidity-particulate-matter-sensor-quick-start-guide) and User Manual (https://ncd.io/blog/industrial-iot-wireless-air-quality-co2-temperature-humidity-particulate-matter-sensor-user-manual).